COER UNIVERSITY
ROORKEE, UTTARAKHAND

Course: BCA

Python Libraries and Applications

Lab Sheet 1

Array Creation & Basics

1. Create a NumPy array of all multiples of 7 between 7 and 100, and print its length.
2. Create a 4×4 array filled entirely with the value 9 without using a loop.
3. Create a 1-D array of 10 elements and convert its data type from integer to float, then verify the change.
4. Create two arrays — one using a Python list and one using `np.array()` with the same values — and compare the memory size of each using `sys.getsizeof()`.

Indexing & Slicing

5. Create a 6×6 array of sequential numbers and extract only the four corner elements using indexing.
6. Given a 1-D array of 12 elements, extract every third element starting from index 1.
7. Create a 5×5 array and use slicing to reverse the order of its rows.
8. Write a program to replace every element divisible by 3 in an array with -1, using Boolean indexing.

Array Operations

9. Write a program to normalize an array (scale all values between 0 and 1) using vectorized operations, without using a loop.
10. Given a 3×3 matrix, compute its transpose and verify that the transpose of the transpose returns the original matrix.
11. Write a program to find the maximum value and its index position in a randomly generated array using `np.argmax()`.
12. Compute the element-wise remainder (modulus) of two integer arrays.

Concatenation & Reshaping

13. Create two 2×3 arrays and stack them to form a 3-D array of shape (2, 2, 3) using `np.stack()`.
14. Write a program to combine three 1-D arrays of unequal starting shapes into a single array using `np.concatenate()`, after first reshaping them appropriately.
15. Take a 1-D array of 16 elements and reshape it into a 4×4 matrix, then use `np.transpose()` to convert it into its column-major equivalent.

Splitting

16. Write a program to split a 3-D array of shape (2, 4, 4) into two equal parts along the first axis.
17. Given an array of 17 elements, use `np.array_split()` to divide it into 4 parts and print the size of each part.

Statistics

18. Write a program to identify and remove outliers from an array (values beyond 2 standard deviations from the mean).
19. Given marks of 5 students in 4 subjects (as a 5×4 array), compute each student's average and each subject's average separately.
20. Compute the correlation coefficient between two 1-D arrays using `np.corrcoef()`.

File I/O

21. Write a program to generate a 5×5 array of random integers, save it as a `.csv` file, then reload it and confirm the loaded array matches the original using `np.array_equal()`.
22. Save three different arrays (of different shapes) into a single `.npz` file, then write a program to load the file and print the names of all arrays stored inside it (using `.files`).

NumPy Random Module

23. Write a program to generate a random 3×3 matrix with values between 10 and 50, then sort each row in ascending order.
24. Simulate flipping a biased coin (70% chance of heads) 200 times using the Binomial distribution and calculate the observed proportion of heads.
25. Generate a random sample of 2,000 values from a Normal distribution with mean 0 and standard deviation 2. Count how many values fall outside the range [-4, 4].
26. Write a program to simulate the number of website visits per minute (lambda = 5) over a 60-minute period using the Poisson distribution, and plot/print the minute with the highest visits.
27. Simulate a survey where respondents choose one of 3 products with probabilities [0.5, 0.3, 0.2]. Use the Multinomial distribution to simulate 200 respondents and print the count for each product.
28. Write a program to generate a random permutation of a deck of 52 cards (represented as numbers 1–52) and deal the first 5 cards as a "hand."